RaceDay - API Endpoint Plan

Programming 2B | Module Code PROG6212 | Part 1 - System Planning and Database

Purpose and design rationale. RaceDay is planned as a resource-oriented REST API. GET, POST, PUT and DELETE follow standard HTTP semantics (IETF, 2022). The resource-based route structure is consistent with the REST architectural style described by Fielding (2000).

The assessment requires two roles: Organiser and Participant. The role requirement is stated explicitly for protected operations. Authentication establishes identity, while authorization determines what an authenticated identity may access; ASP.NET Core supports role-based authorization (Microsoft, 2026a).

The endpoint set covers authentication, user profile, events, categories, event enrolments and results. EventCategories is included because event-category associations need their own data, including entry fee and participant limits.

Endpoint specification

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Creates a new Participant account. Public registration does not allow a caller to assign an Organiser role.	Public	firstName, lastName, email, password, phoneNumber (optional)	201 Created; 400 Bad Request; 409 Conflict
POST	/api/auth/login	Authenticates a registered user and returns an access token for protected requests.	Public	email, password	200 OK; 400 Bad Request; 401 Unauthorized
GET	/api/users/me	Returns the authenticated user's own profile.	Authenticated	None	200 OK; 401 Unauthorized
PUT	/api/users/me	Updates the authenticated user's own profile.	Authenticated	firstName, lastName, phoneNumber	200 OK; 400 Bad Request; 401 Unauthorized
GET	/api/events	Returns available RaceDay events for browsing.	Public	None	200 OK
GET	/api/events/{id}	Returns details for one RaceDay event.	Public	None	200 OK; 404 Not Found
POST	/api/events	Creates an event owned by the authenticated Organiser.	Organiser	name, description, eventDate, location, registrationDeadline, imageUrl (optional), routeInfoUrl (optional)	201 Created; 400 Bad Request; 401 Unauthorized; 403 Forbidden
PUT	/api/events/{id}	Updates an event managed by the authenticated Organiser.	Organiser	name, description, eventDate, location, registrationDeadline, imageUrl (optional), routeInfoUrl (optional)	200 OK; 400 Bad Request; 403 Forbidden; 404 Not Found
DELETE	/api/events/{id}	Deletes an event managed by the authenticated Organiser when related enrolments do not prevent deletion.	Organiser	None	204 No Content; 403 Forbidden; 404 Not Found; 409 Conflict
GET	/api/categories	Returns the available event categories.	Public	None	200 OK
GET	/api/categories/{id}	Returns one event category.	Public	None	200 OK; 404 Not Found
POST	/api/categories	Creates a reusable event category.	Organiser	name, description (optional)	201 Created; 400 Bad Request; 403 Forbidden; 409 Conflict
PUT	/api/categories/{id}	Updates an existing event category.	Organiser	name, description (optional)	200 OK; 400 Bad Request; 403 Forbidden; 404 Not Found; 409 Conflict
DELETE	/api/categories/{id}	Deletes a category that is not currently associated with an event.	Organiser	None	204 No Content; 403 Forbidden; 404 Not Found; 409 Conflict
GET	/api/events/{eventId}/categories	Returns categories configured for an event.	Public	None	200 OK; 404 Not Found
POST	/api/events/{eventId}/categories	Associates a category with an event and defines entry rules.	Organiser	categoryId, entryFee, maximumParticipants (optional)	201 Created; 400 Bad Request; 403 Forbidden; 404 Not Found; 409 Conflict
PUT	/api/events/{eventId}/categories/{id}	Updates the event-specific category configuration.	Organiser	entryFee, maximumParticipants (optional)	200 OK; 400 Bad Request; 403 Forbidden; 404 Not Found
DELETE	/api/events/{eventId}/categories/{id}	Removes a category association when no enrolments depend on it.	Organiser	None	204 No Content; 403 Forbidden; 404 Not Found; 409 Conflict
POST	/api/enrolments	Enrols the authenticated Participant in an event category.	Participant	eventCategoryId	201 Created; 400 Bad Request; 401 Unauthorized; 403 Forbidden; 404 Not Found; 409 Conflict
GET	/api/enrolments/my	Returns only the authenticated Participant's own enrolments.	Participant	None	200 OK; 401 Unauthorized; 403 Forbidden
GET	/api/events/{eventId}/enrolments	Returns all enrolments for an event managed by the authenticated Organiser.	Organiser	None	200 OK; 403 Forbidden; 404 Not Found
POST	/api/results	Records a result against a participant enrolment.	Organiser	enrolmentId, finishPosition (optional), finishTime (optional), resultStatus	201 Created; 400 Bad Request; 403 Forbidden; 404 Not Found; 409 Conflict
GET	/api/results/my	Returns the authenticated Participant's personal results.	Participant	None	200 OK; 401 Unauthorized; 403 Forbidden
PUT	/api/results/{id}	Updates a previously recorded result.	Organiser	finishPosition (optional), finishTime (optional), resultStatus	200 OK; 400 Bad Request; 403 Forbidden; 404 Not Found

Design consistency and references

The endpoint plan is intended to map directly to the six-entity relational model: Users, Events, Categories, EventCategories, Enrolments and Results. Primary and foreign keys support entity and referential integrity, while UNIQUE and CHECK constraints support data integrity (Microsoft, 2026b; Microsoft, 2026c).

The ERD and SQL use the same entity and relationship names. Public registration creates a Participant account; Organiser access is treated as a protected role. Dynamic weather information is treated as external data, while an optional RouteInfoUrl field can point users to route information.

References

Fielding, R.T. (2000) *Architectural Styles and the Design of Network-based Software Architectures*. PhD dissertation. University of California, Irvine. Available at: https://ics.uci.edu/~fielding/pubs/dissertation/ (Accessed: 22 September 2026).

IETF (2022) *RFC 9110: HTTP Semantics*. Available at: https://www.rfc-editor.org/rfc/rfc9110.html (Accessed: 22 September 2026).

Microsoft (2026a) *Introduction to authorization in ASP.NET Core*. Microsoft Learn. Available at: https://learn.microsoft.com/en-us/aspnet/core/security/authorization/introduction (Accessed: 22 September 2026).

Microsoft (2026b) *Primary and foreign key constraints - SQL Server*. Microsoft Learn. Available at: https://learn.microsoft.com/en-us/sql/relational-databases/tables/primary-and-foreign-key-constraints (Accessed: 22 September 2026).

Microsoft (2026c) *Unique constraints and check constraints - SQL Server*. Microsoft Learn. Available at: https://learn.microsoft.com/en-us/sql/relational-databases/tables/unique-constraints-and-check-constraints (Accessed: 22 September 2026).

Source note: The ERD is the author's own compilation from the RaceDay assessment requirements.